

ANISH KHURANA

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EDUCATION

University of Waterloo

Sep 2021 – Apr 2026

Bachelor of Software Engineering

PROFESSIONAL EXPERIENCE

University of Waterloo – Undergraduate Research Assistant

Jan 2026 – Apr 2026

Supervisor: Dr. Gennaro Notomista

- Enabled concurrent multi-robot orchestration by designing a ROS2 management platform integrating live UDP discovery, automated session management, and a centralized React dashboard.
- Built a real-time WebSocket telemetry system streaming **25+ ROS2 topics** for live robot monitoring and control.
- Strengthened distributed system reliability by diagnosing and resolving Docker networking and DDS middleware failures.

Zoocasa – QA Automation Developer

Sep 2025 – Dec 2025

- Increased automated test coverage by **40%** by integrating Jest and React Testing Library suites into GitHub Actions CI for push and pull-request validation.
- Improved data integrity in a multi-tenant real estate platform by delivering production features across React/Next.js and Ruby on Rails/Go services.
- Identified **50+** defects through end-to-end release testing across staging and production, reducing the risk of customer-facing regressions.
- Built an AI prototype that classified listing photos and generated 360° virtual property tours using Stable Diffusion.

Microart Services Inc – Software Engineering Intern

Sep 2023 – Dec 2023 | Jan 2025 – Apr 2025

- Reduced production data retrieval latency by **33%** by redesigning the data access layer with PostgreSQL/Supabase remote procedure calls (RPCs) for high-frequency operations.
- Delivered an end-to-end user review feature across React Native, Supabase, and PostgreSQL, **reducing review submission time by 30%**.
- Implemented Bluetooth Low Energy (BLE) over-the-air (OTA) firmware update workflows, enabling 100% successful updates for supported production devices.
- Automated large-scale data ingestion by building a JavaScript pipeline that scraped, transformed, and loaded **100K+ records** into PostgreSQL.

VCT Group – Software Engineering Associate

May 2024 – Aug 2024

- Modernized a major production application module using React, TypeScript, and Spring Boot, improving maintainability.
- Improved full-stack maintainability by redesigning frontend-backend interactions using domain-driven design principles.
- Accelerated developer onboarding by co-authoring system architecture and data flow diagrams.

Okos Smart Homes - Software Development Intern

Jan 2023 - Apr 2023

- Reduced API response latency by **95%** by redesigning REST endpoints and optimizing request/response schemas, eliminating production request failures caused by oversized payloads.
- Developed internal applications supporting daily operations for **20+ project managers** and **200+ field technicians**.
- Delivered production features using React-Admin and Flask, including status visualization and secure account recovery.

PROJECTS

AccessWise - Final Year Capstone Design Project (github.com/programmer812/AccessWise)

- Built a full-stack accessibility platform that automated campus accessibility evaluation using computer vision.
- Developed ML-powered backend pipelines integrating public datasets and user feedback for AODA-aligned accessibility mapping.

Math Notes to Latex Converter (github.com/rohanvedula/PaZaR)

- Built a Raspberry Pi pipeline converting handwritten math notes into LaTeX PDFs using OpenCV and OCR.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, Go, C/C++, SQL, C#, HTML/CSS

Frameworks/Systems: React, Next.js, React Native, Node.js, FastAPI, Flask, Rails, GraphQL, Docker, Git, ROS2

Databases/Cloud: PostgreSQL, MongoDB, Firebase, Supabase, AWS, GCP

Testing: Jest, Playwright, Pytest, React Testing Library

AI/ML: TensorFlow, PyTorch, Scikit-learn